DS-A4 | MODEL COMPARISON & CALIBRATION LAB

Machine-learning prediction of disagreement between GBPQ and GASP barometric estimates

2,084	11	3	5
observations	mineral predictors	classifiers	source-grouped folds

ABSTRACT

This study evaluates whether mineral-composition variables can predict disagreement between GBPQ and GASP pressure estimates. Logistic Regression, Random Forest and Gradient Boosting were evaluated using five-fold source-grouped cross-validation. Performance was assessed using precision-recall AUC, ROC-AUC, Brier score and calibration diagnostics; threshold-cost sensitivity, error slices and source-cluster bootstrap comparisons were also examined.

Main finding. Random Forest achieved the highest PR-AUC for Mg_10 (0.2359), while Gradient Boosting achieved the highest PR-AUC for Fe_10 (0.2378). Paired bootstrap comparisons did not establish clear differences between the two tree-based models.

1. Research objectives

The objective was to assess whether mineral compositions contain predictive information about substantial disagreement between GBPQ and GASP barometric estimates. GASP is treated as a comparator rather than unquestionable geological truth.

2. Dataset and predictors

The modelling dataset contains 2,084 observations and 11 predictors: Fe_Bt, Mg_Bt, AlVI_Bt, Ti_Bt, Ca_Pl, Na_Pl, K_Pl, Fe_Grt, Mg_Grt, Ca_Grt and Mn_Grt. Positive cases were rare: 96 Mg_10 cases (4.61%) and 45 Fe_10 cases (2.16%). Source publication was used to group observations during cross-validation.

Opaque assemblage information was unreported for 1,130 observations (54.22%). Missing information was not interpreted as evidence that opaque minerals were absent.

3. Modelling and validation

Three fixed classifiers were compared: Logistic Regression with standardisation, Random Forest and Gradient Boosting. All models used the same five source-grouped folds, and out-of-fold predictions were used for evaluation. Metrics included PR-AUC, ROC-AUC, Brier score, ten-bin expected calibration error, precision-recall curves, ROC curves and calibration curves. No hyperparameter optimisation was performed.

Validation boundary: source-grouped cross-validation limits publication-level leakage, but independent external validation is still required. The observed class imbalance makes PR-AUC particularly important when interpreting discrimination.

4. Model comparison and diagnostics

The table reports out-of-fold performance. Higher PR-AUC and ROC-AUC indicate stronger discrimination; lower Brier score indicates better probabilistic accuracy. ECE is a ten-bin calibration summary.

Target	Model	PR-AUC	ROC-AUC	Brier	ECE
Mg_10	Logistic Regression	0.1109	0.7620	0.0438	0.0100
Mg_10	Random Forest	0.2359	0.8631	0.0389	0.0088
Mg_10	Gradient Boosting	0.2218	0.8031	0.0396	0.0067
Fe_10	Logistic Regression	0.1024	0.8376	0.0209	0.0044
Fe_10	Random Forest	0.2090	0.8764	0.0190	0.0034
Fe_10	Gradient Boosting	0.2378	0.7812	0.0188	0.0070

Mg_10: Random Forest has the strongest point estimates across PR-AUC, ROC-AUC and Brier score. **Fe_10:** Gradient Boosting has the highest PR-AUC and lowest Brier score, while Random Forest has the higher ROC-AUC.

Out-of-fold diagnostics — Mg_10

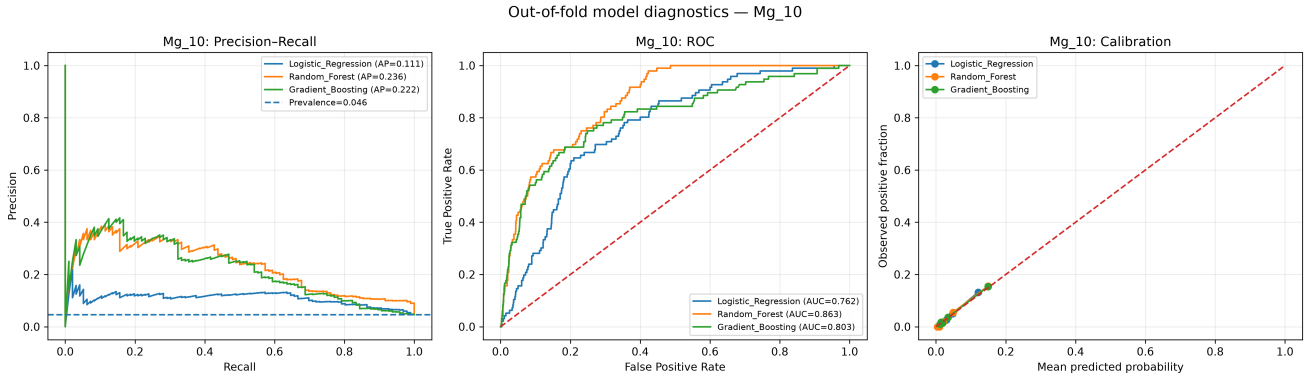


Figure 1. Precision-recall, ROC and calibration diagnostics for Mg_10.

Out-of-fold diagnostics — Fe_10

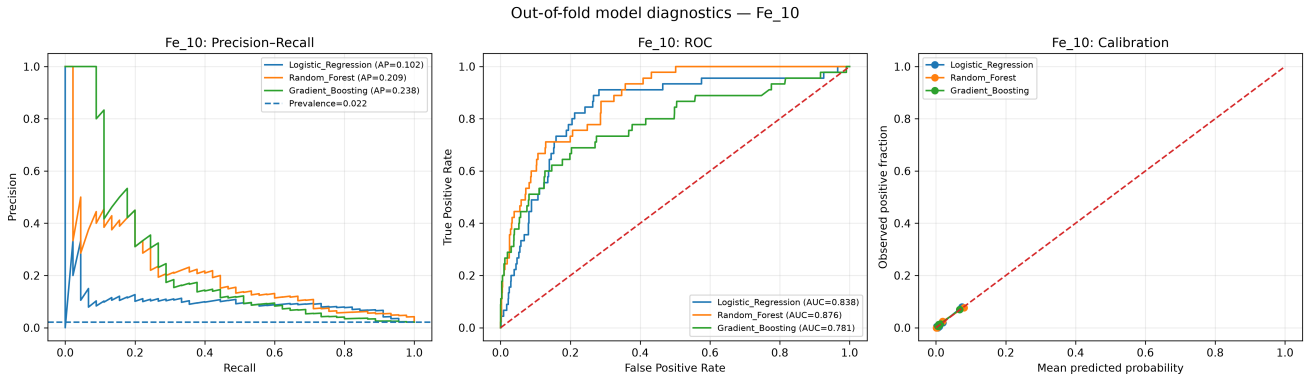


Figure 2. Precision-recall, ROC and calibration diagnostics for Fe_10.

5. Paired source-cluster bootstrap

A paired source-cluster bootstrap with 1,000 valid replicates compared model metrics. Differences are Model A minus Model B. For PR-AUC, positive values favour Model A; for Brier score, negative values favour Model A. Confidence intervals crossing zero do not establish a clear paired difference.

Target	Model A	Model B	Metric	Difference	95% CI
Fe_10	Gradient Boosting	Logistic Regression	PR-AUC	0.1299	[0.0064, 0.2596]
Fe_10	Gradient Boosting	Logistic Regression	Brier	-0.0021	[-0.0041, -0.0002]
Fe_10	Gradient Boosting	Random Forest	PR-AUC	0.0231	[-0.0465, 0.0951]
Fe_10	Gradient Boosting	Random Forest	Brier	-0.0003	[-0.0013, 0.0007]
Fe_10	Logistic Regression	Random Forest	PR-AUC	-0.1068	[-0.2154, -0.0035]
Fe_10	Logistic Regression	Random Forest	Brier	0.0019	[0.0005, 0.0033]
Mg_10	Gradient Boosting	Logistic Regression	PR-AUC	0.1149	[0.0427, 0.1941]
Mg_10	Gradient Boosting	Logistic Regression	Brier	-0.0043	[-0.0074, -0.0015]
Mg_10	Gradient Boosting	Random Forest	PR-AUC	-0.0141	[-0.0595, 0.0328]
Mg_10	Gradient Boosting	Random Forest	Brier	0.0007	[-0.0008, 0.0022]
Mg_10	Logistic Regression	Random Forest	PR-AUC	-0.1289	[-0.2079, -0.0550]
Mg_10	Logistic Regression	Random Forest	Brier	0.0050	[0.0026, 0.0077]

Interpretation

For both targets, paired intervals comparing Random Forest with Gradient Boosting include zero for PR-AUC and Brier score. Their relative superiority is therefore not established by this bootstrap. The tree-based models compare favourably with Logistic Regression on several metrics, although inference remains constrained by the small number of positive cases.

The bootstrap quantifies uncertainty under source-level resampling. It does not prove equivalence between models and does not constitute external validation. The number and diversity of independent source publications remain important.

6. Threshold and cost analysis

An illustrative false-negative to false-positive cost ratio of 5:1 was examined. The selected thresholds below are exploratory because they were optimised on pooled out-of-fold predictions.

Target	Model	Threshold	Review rate	Precision	Recall
Mg_10	Random Forest	0.1719	6.29%	31.30%	42.71%
Fe_10	Gradient Boosting	0.1850	1.49%	35.48%	24.44%

Decision caveat. The 5:1 cost ratio is an assumed scenario, not a measured operational consequence. Thresholds selected from pooled out-of-fold predictions may have optimistic operating characteristics.

Mg_10 — cost versus review rate

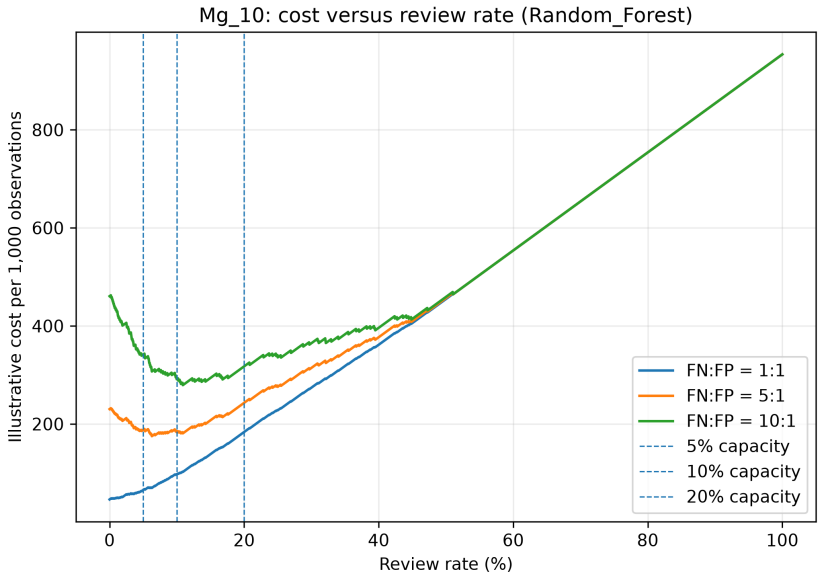


Figure 3. Illustrative cost per 1,000 observations against review rate for Mg_10.

Fe_10 — cost versus review rate

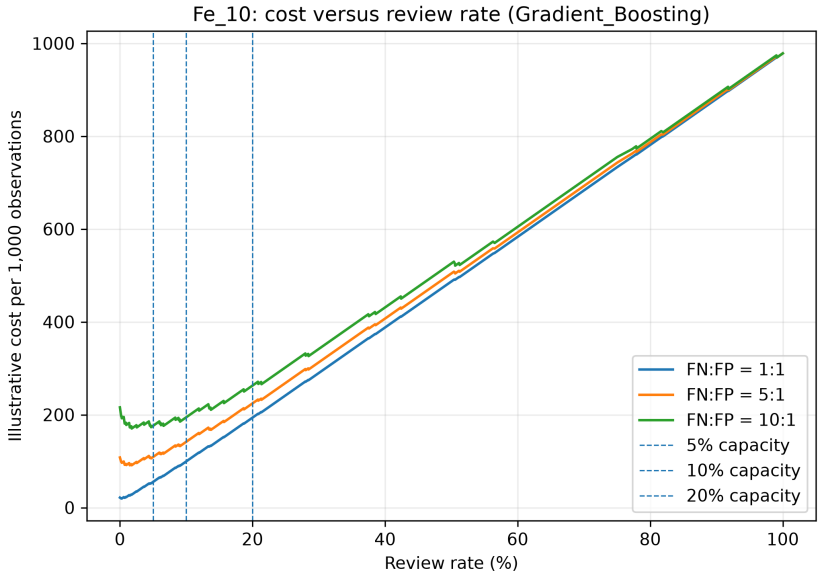


Figure 4. Illustrative cost per 1,000 observations against review rate for Fe_10.

7. Error analysis

Most observations were metapelites (2,013 of 2,084). For Mg_10, Random Forest missed 55 positive cases at the provisional threshold. For Fe_10, Gradient Boosting missed 34 positive cases. These false negatives demonstrate that the models cannot reliably identify every disagreement.

Other rock-type groups were small, and subgroup performance estimates are consequently unstable. Error slices should be interpreted descriptively rather than as evidence of reliable geological differences.

8. Limitations

- Rare positive cases: 96 Mg_10 cases (4.61%) and 45 Fe_10 cases (2.16%).
- GASP is a comparator rather than independently verified ground truth.
- Opaque assemblage information is unreported for 54.22% of observations.
- Uneven and small rock-type subgroups limit subgroup-level inference.
- Illustrative cost ratios do not represent measured operational consequences.
- Threshold optimisation on pooled out-of-fold predictions may produce optimistic estimates.
- Independent external validation remains necessary.

9. Conclusion

Mineral-composition predictors contain some signal for identifying GBPQ–GASP disagreement. Random Forest is retained as the provisional Mg_10 model and Gradient Boosting as the provisional Fe_10 model. However, paired bootstrap comparisons do not establish clear superiority between the two tree-based alternatives. The models should support further investigation rather than replace experimental calibration or geological judgement.

10. Reproducibility

The project audit records Python 3.12.7, pandas 3.0.6, NumPy 2.4.6, scikit-learn 1.9.1 and Matplotlib 3.11.1. SHA-256 hashes are recorded for principal analysis outputs. The audit confirms the presence of listed files; it does not by itself demonstrate a clean end-to-end rerun.

Prepared from the DS-A4 project analysis outputs. Numerical findings and figures were carried forward; no new model fitting or independent validation was performed during report generation.